

# Global Air Connectivity and Trade Openness, 1996–2023

*Building a global aviation connectivity index, and estimating its effect on trade*

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# Motivation

- We simply wanted to Graduate from Railway research.
- We had a chance to meet Professor Anming Zhang (Chief editor of Transportation Research Part A)
- He suggested me about aviation research.

# What this paper does

- We build a new measure of ***how connected each country is to the world by air***, covering 184 countries from 1996 to 2023.
- We then ask whether that connectivity causally raises a country's **trade openness**, using an instrument based on tourism and UNESCO national heritage to break reverse causality.

**A data contribution (the index) and a causal contribution (the effect on trade).**

# Five parts

**01**

## **Motivation**

Why connectivity & trade

**02**

## **Building GACI**

What it is and how we  
build it

**03**

## **Identification**

Tourism–heritage IV

**04**

## **Results**

Effect, who gains, how  
much

**05**

## **Conclusion**

Policy & extensions



# 01

## Motivation

*Why air connectivity, and why trade*

# Air connectivity shrinks economic distance

- A country's position in the world air network sets how near it is, in time and cost, to markets, suppliers, and partners.
- Among the flows distance impedes (people, capital, ideas, goods), trade responds most robustly and is observed most cleanly.
- Trade is the proximate, observable margin on which connectivity acts; income is the distant, confounded end of the chain.

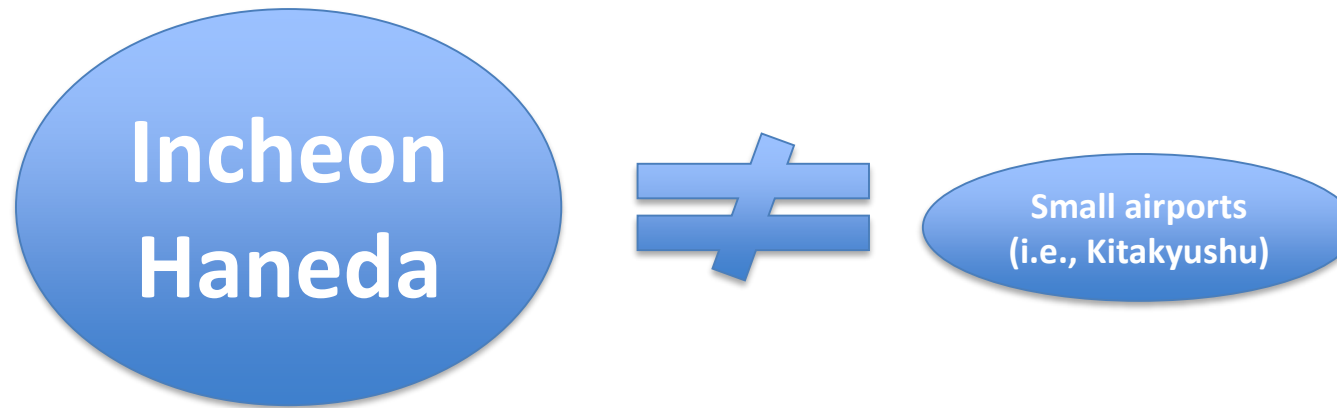
**So we study trade: the clearest economic footprint of air access.**

# Why connectivity should move trade: two channels

**Why a continuous index, not a count or a discrete treatment-** Continuous, not lumpy.

Airports rarely open or close; position moves every year as partners rewire. A discrete DID misses this.

**Weighted, not counted.** A mega-hub  $\neq$  a thin regional field (top ~2.4% of airports carry half the world's seats). A count erases this; a capacity-weighted index keeps it.



- **Captures shocks.** War, sanctions, COVID collapse connectivity while airports stay open. No open/close event to see but the index registers the drop.



# 02

## Building GACI

*What is it, and how do we construct it?*

# What is "air connectivity", really?

- It is not how many airports or routes a country has.
- **It is a country's POSITION in the world air network: how easily, through how many and how important partners, it can reach everywhere else.**
- Two countries with the same number of routes differ if one connects to global hubs and the other to dead-ends.

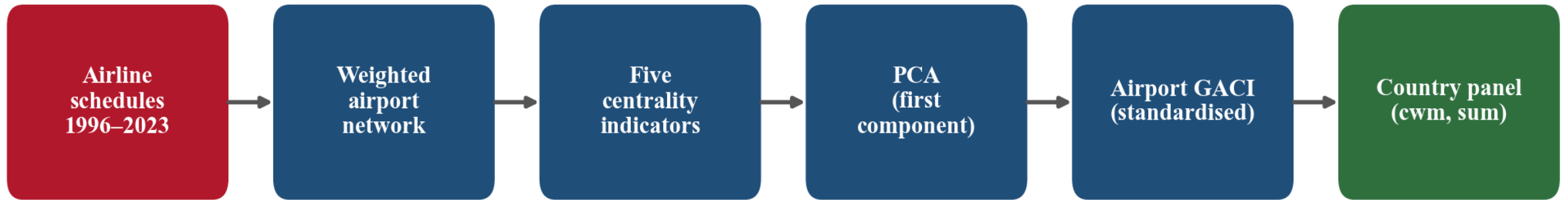
**Connectivity = network position, not a simple count.**

# How we measure it: the Global Air Connectivity Index

- GACI summarises each airport's position in the weighted world air network as a single, time-comparable number (Cheung, Wong & Zhang, 2020).
- **It blends three ideas:**
  - REACH — how many and how well-placed are its connections;
  - FLOW — how much passenger traffic moves through it;
  - STANDING — how important are the airports it connects to.

**One number that captures reach, flow, and standing in the global network.**

# How the index is built: the pipeline



*BUILD THE INDEX (this paper's data contribution)*

*USE FOR TRADE*

*Each step is computed for every year, then aggregated to countries.*

## 元データ

| CarrDom1 | CarrDom1Name                | Origin | Origin.1        | DepCity | DepCityName     | DeplATACT | DeplATACT  | FstSeats | BusSeats | EcoSeats | Frequency | Seats |
|----------|-----------------------------|--------|-----------------|---------|-----------------|-----------|------------|----------|----------|----------|-----------|-------|
| US       | USA                         | LAX    | Los Angeles     | LAX     | Los Angeles     | US        | USA        | 0        | 0        | 175      | 1         | 175   |
| NO       | Norway                      | TOS    | Tromso          | TOS     | Tromso          | NO        | Norway     | 0        | 0        | 39       | 1         | 39    |
| US       | USA                         | SAT    | San Antonio     | SAT     | San Antonio     | US        | USA        | 0        | 12       | 114      | 1         | 126   |
| CN       | China                       | CTU    | Chengdu (CTU)   | CTU     | Chengdu (CTU)   | CN        | China      | 0        | 30       | 271      | 1         | 301   |
| US       | USA                         | BOG    | Bogota          | BOG     | Bogota          | CO        | Colombia   | 0        | 0        | 0        | 1         | 0     |
| NZ       | New Zealand                 | PER    | Perth           | PER     | Perth           | AU        | Australia  | 0        | 12       | 162      | 1         | 174   |
| CN       | China                       | SZX    | Shenzhen (SZX)  | SZX     | Shenzhen (SZX)  | CN        | China      | 8        | 0        | 153      | 1         | 161   |
| IN       | India                       | FRA    | Frankfurt (FRA) | FRA     | Frankfurt (FRA) | DE        | Germany    | 0        | 24       | 144      | 1         | 168   |
| DE       | Germany                     | LYS    | Lyon St-ex      | LYS     | Lyon            | FR        | France     | 0        | 28       | 126      | 1         | 154   |
| US       | USA                         | SAN    | San Diego       | SAN     | San Diego       | US        | USA        | 0        | 16       | 144      | 1         | 160   |
| US       | USA                         | PHL    | Philadelphia    | PHL     | Philadelphia    | US        | USA        | 0        | 20       | 170      | 1         | 190   |
| MX       | Mexico                      | QRO    | Queretaro       | QRO     | Queretaro       | MX        | Mexico     | 0        | 0        | 186      | 1         | 186   |
| TZ       | Tanzania United Republic of | SEU    | Seronera        | SEU     | Seronera        | TZ        | Tanzania U | 0        | 0        | 13       | 1         | 13    |
| US       | USA                         | DAL    | Dallas Love     | DFW     | Dallas          | US        | USA        | 0        | 0        | 143      | 1         | 143   |
| US       | USA                         | BNE    | Brisbane        | BNE     | Brisbane        | AU        | Australia  | 0        | 8        | 168      | 1         | 176   |
| QA       | Qatar                       | CKY    | Conakry         | CKY     | Conakry         | GN        | Guinea     | 0        | 12       | 147      | 1         | 159   |
| CN       | China                       | PKX    | Beijing Dax     | BJS     | Beijing         | CN        | China      | 0        | 12       | 147      | 1         | 159   |
| CN       | China                       | CAN    | Guangzhou       | CAN     | Guangzhou       | CN        | China      | 0        | 17       | 162      | 1         | 179   |
| CN       | China                       | XIY    | Xi'an Xian      | SIA     | Xi'an           | CN        | China      | 0        | 0        | 0        | 1         | 0     |
| GB       | United Kingdom              | MIA    | Miami Inter     | MIA     | Miami           | US        | USA        | 0        | 20       | 214      | 1         | 234   |
| KR       | Korea Republic of           | YHZ    | Halifax Star    | YHZ     | Halifax         | CA        | Canada     | 0        | 12       | 124      | 1         | 136   |
| US       | USA                         | LIR    | Guanacaste      | LIR     | Guanacaste      | CR        | Costa Rica | 0        | 8        | 120      | 1         | 128   |
| ES       | Spain                       | LPA    | Gran Canaria    | LPA     | Gran Canaria    | ES        | Spain      | 0        | 0        | 70       | 1         | 70    |
| QA       | Qatar                       | RDU    | Raleigh/Dur     | RDU     | Raleigh/Dur     | US        | USA        | 0        | 12       | 64       | 1         | 76    |
| US       | USA                         | HND    | Tokyo Intl      | TYO     | Tokyo           | JP        | Japan      | 5        | 42       | 205      | 1         | 252   |

Connected Dummy

Total Riding capacity

## Step 0: building a clean weighted network

- Source: global airline schedule data (route-segment level). OAG Datasource (So big!)
- Keep only scheduled **commercial service**; exclude military, charter, seaplane, heliports, and airports that closed in-window.
- Aggregate segments per airport pair per year into one link; set the link weight to passenger seat capacity.
- **Two matrices per year: adjacency A (who connects) and weight W (how much).**

|     | ATL | LHR | DXB | HKG | SPK |
|-----|-----|-----|-----|-----|-----|
| ATL | —   | 1   | 1   | 1   | 1   |
| LHR | 1   | —   | 1   | 1   | 0   |
| DXB | 1   | 1   | —   | 1   | 0   |
| HKG | 1   | 1   | 1   | —   | 0   |
| SPK | 1   | 0   | 0   | 0   | —   |

|     | ATL | LHR | DXB | HKG | SPK |
|-----|-----|-----|-----|-----|-----|
| ATL | —   | 9.0 | 5.0 | 4.0 | 0.3 |
| LHR | 9.0 | —   | 6.0 | 5.0 | 0   |
| DXB | 5.0 | 6.0 | —   | 7.0 | 0   |
| HKG | 4.0 | 5.0 | 7.0 | —   | 0   |
| SPK | 0.3 | 0   | 0   | 0   | —   |

# Step 1a: topological indicators (position)

- Degree — number of direct connections (raw reach).
- Closeness — inverse of the average number of **Transfers** to reach all others (accessibility).

Hop matrix (shortest number of connections)

|     | ATL | HND | NRT | FUK | Row average |
|-----|-----|-----|-----|-----|-------------|
| ATL | –   | 1   | 1   | 2   | 1.33        |
| HND | 1   | –   | 1   | 1   | 1.00        |
| NRT | 1   | 1   | –   | 2   | 1.33        |
| FUK | 2   | 1   | 2   | –   | 1.67        |

(e.g., FUK→ATL is FUK-HND-ATL = 2; FUK→NRT is FUK-HND-NRT = 2)

Closeness = 1 ÷ row average

| Airport | Avg. hops | Closeness | Interpretation                                |
|---------|-----------|-----------|-----------------------------------------------|
| HND     | 1.00      | 1.00      | Hub – reaches everyone in 1 hop, most central |
| ATL     | 1.33      | 0.75      | Intermediate                                  |
| NRT     | 1.33      | 0.75      | Intermediate                                  |
| FUK     | 1.67      | 0.60      | Spoke – always via HND, most peripheral       |

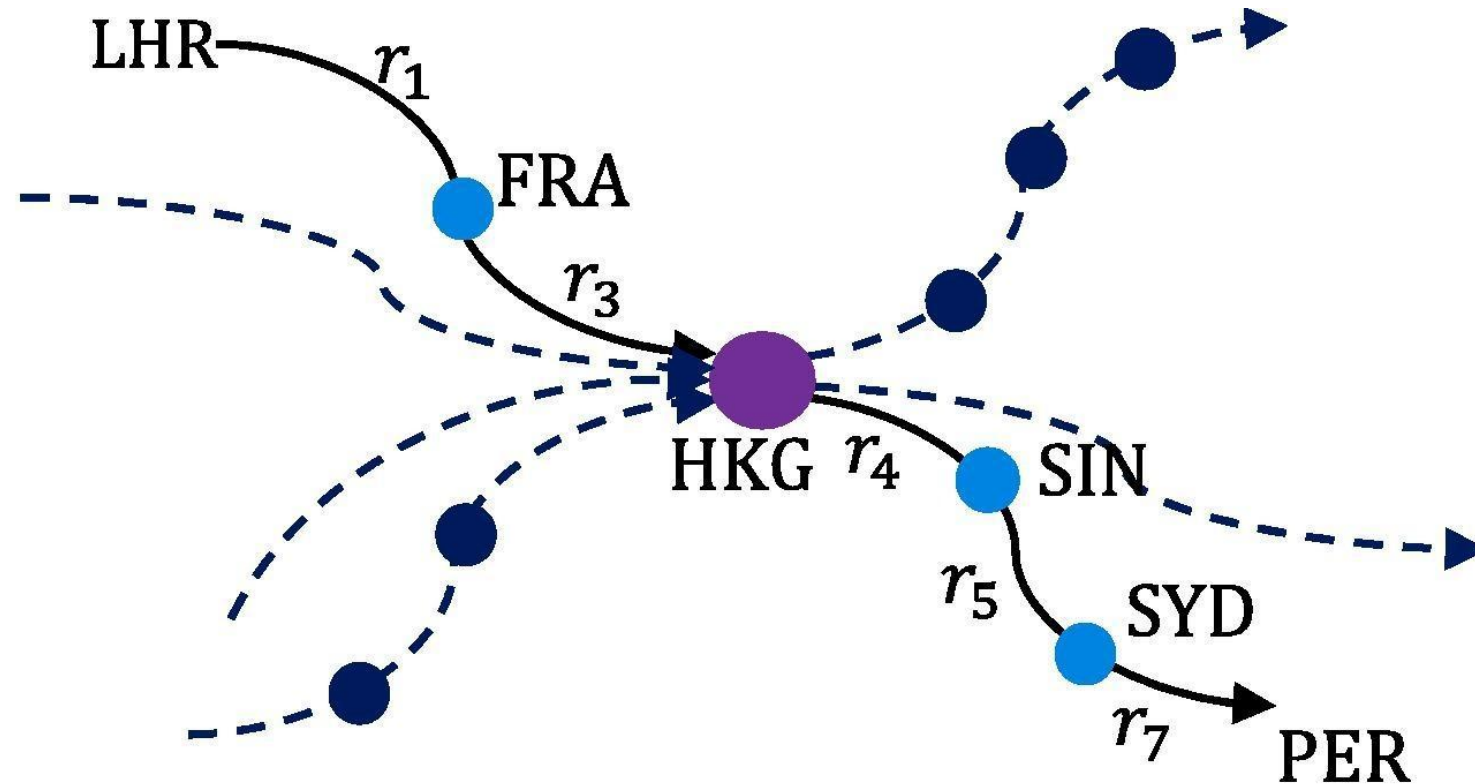
+ Eigenvector:  
How much I am connected to “BIG” airports.

| Airport | Degree (# friends) | Eigenvector (rank of friends) |
|---------|--------------------|-------------------------------|
| HND     | 3                  | High                          |
| ATL     | 2                  | High                          |
| NRT     | 2                  | High                          |
| FUK     | 1                  | Low                           |

## Step 1b: volumetric indicators (passenger flow)

- Flow betweenness — **share of passenger transferring through the airport; the hub / transfer role. (Haneda < Narita) .**
- Regional importance — **average connection intensity within its own region (Haneda > Narita) .**
- **These two flow measures make GACI flow-aware, not purely topological.**

# Flow betweenness: passengers as electrical current



*Each route is a resistor with resistance inversely proportional to its passenger intensity; betweenness = current through the node. Source: Cheung et al. (2020).*

# Step 1: the exact formulas

## Step 1: five airport-level centrality indicators

*Built on the weighted airport network  $G(V, E)$ , link weight  $w_{ij}$  = passenger seat capacity*

### Degree

$$d_i = \sum_j a_{ij} \quad (\text{number of direct connections})$$

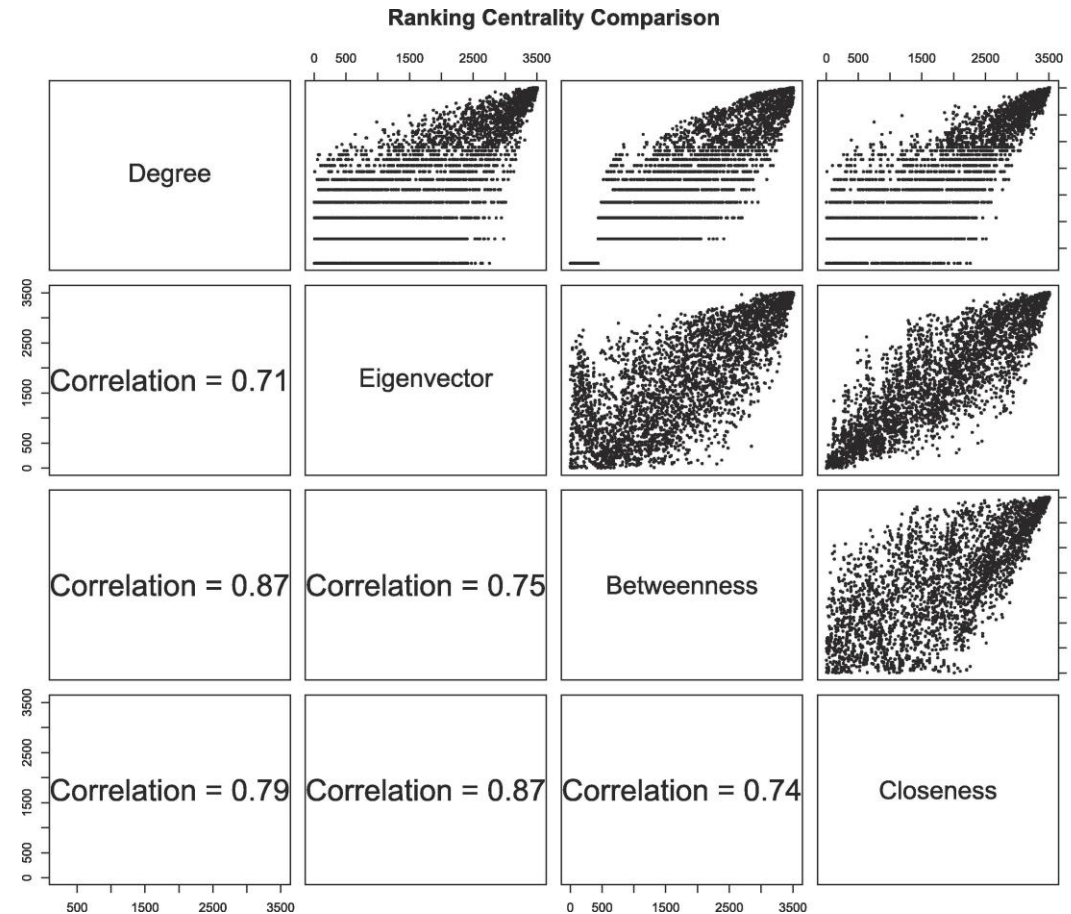
### Flow betweenness (hub / transfer role)

$$F_b(i) = \frac{1}{N_B} \sum_{j \neq g} \tau_{jg}(i), \quad \text{resistance} \propto 1/\bar{w}_{ij}, \quad \bar{w}_{ij} = \sqrt{(s_i/d_i)(s_j/d_j)}$$

### Closeness, Eigenvector, Regional importance

$$C_c(i) = \frac{N-1}{\sum_j \text{Dist}(i,j)}, \quad C_E(i) = \frac{1}{\lambda} \sum_j W_{ij} C_E(j), \quad C_R(i) = \bar{w}_{ij}^{\text{region}}$$

# The five indicators are related but not redundant



Pairwise rank correlations 0.71–0.87: each adds information, so we combine rather than pick one. Source: Cheung et al. (2020).

## Step 2: combine the five into one index (PCA)

- Five indicators, different scales, partly correlated.
- **GACI = the first principal component: the single weighted combination capturing the most variance across airports.**
- It explains about two-thirds of total variance, so one number retains most of the information.

## Step 2: the aggregation formula

### Step 2: combine into one index by PCA

*Each indicator is standardised (comparable within year and across years)*

#### Composite index

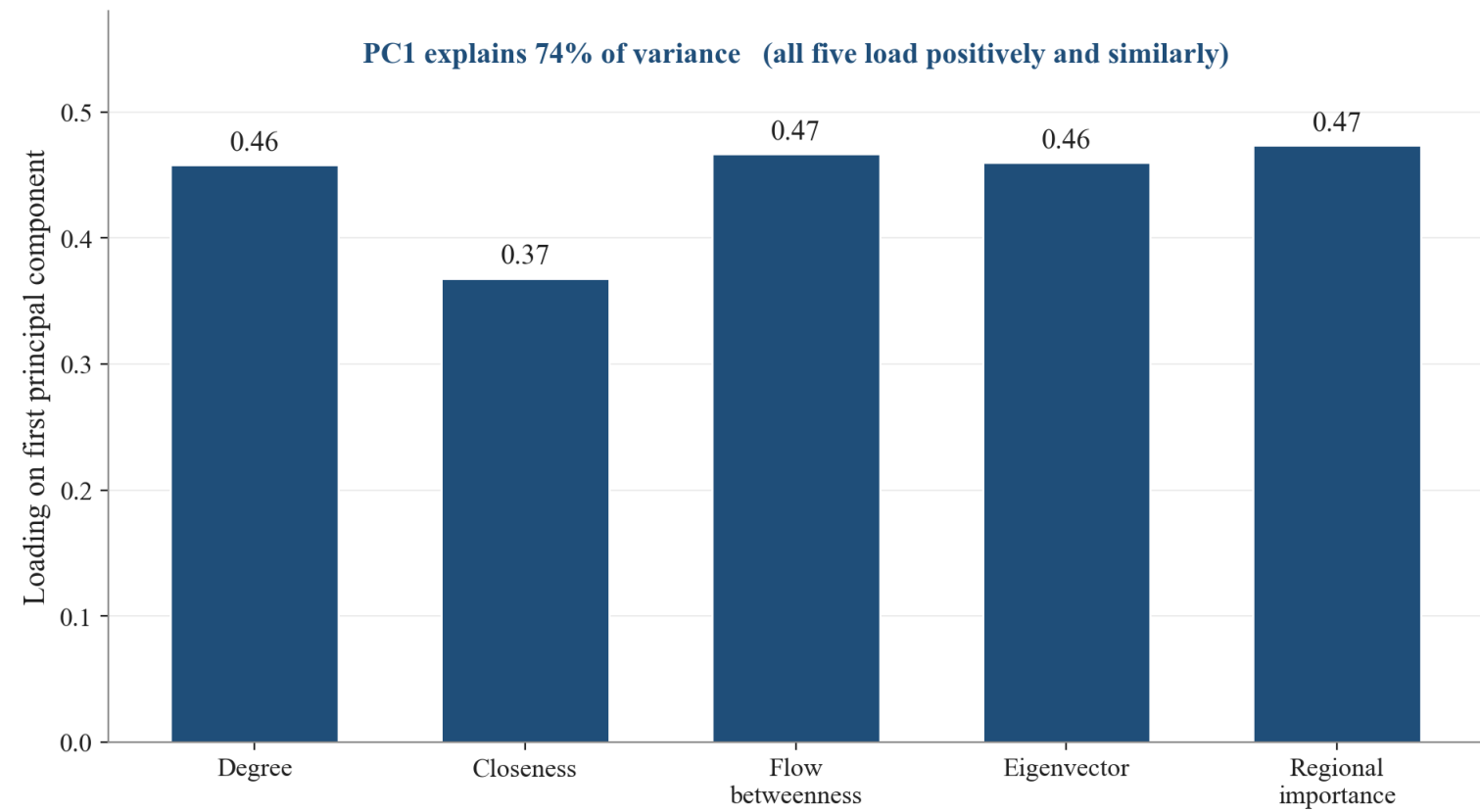
$$\text{GACI} = [D, C_c, F_b, C_E, C_R] \omega$$

$\omega$  = first principal component (direction of maximum variance)

*The first component explains 65--68% of total variance*

*Result: an airport-level, time-comparable connectivity score*

# PCA weights, computed on our panel

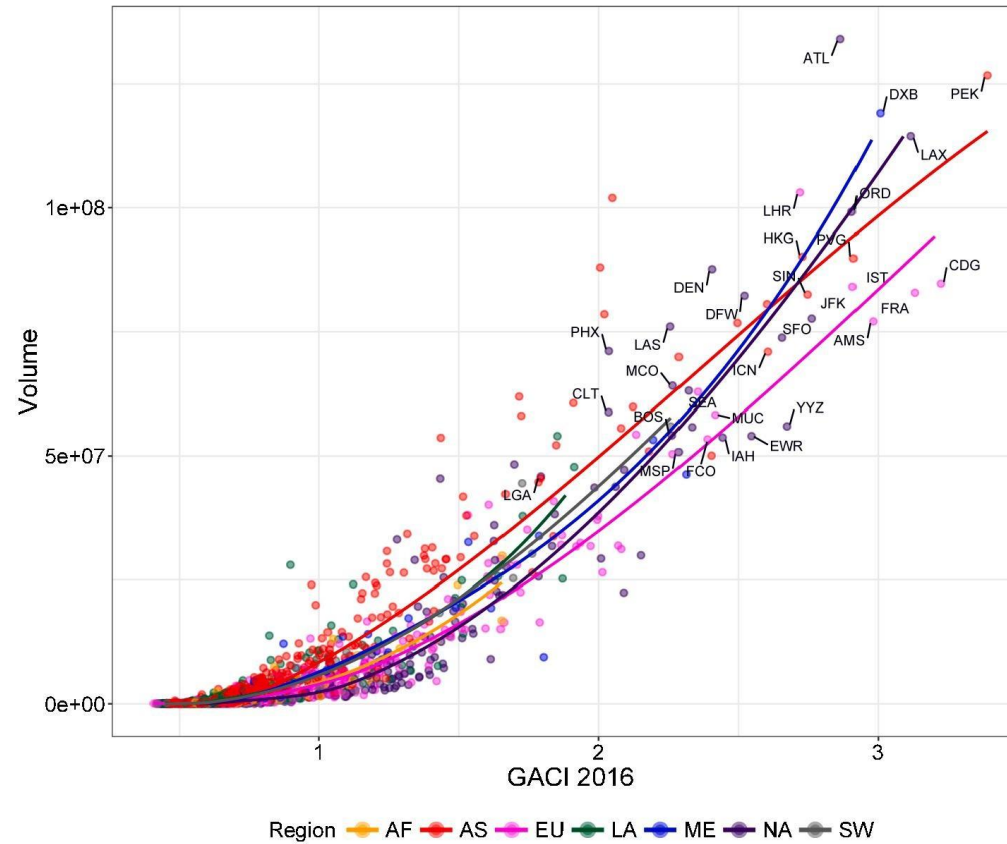


First-component loadings on our 1996–2023 data: all positive and similar; PC1 explains 74% of variance.

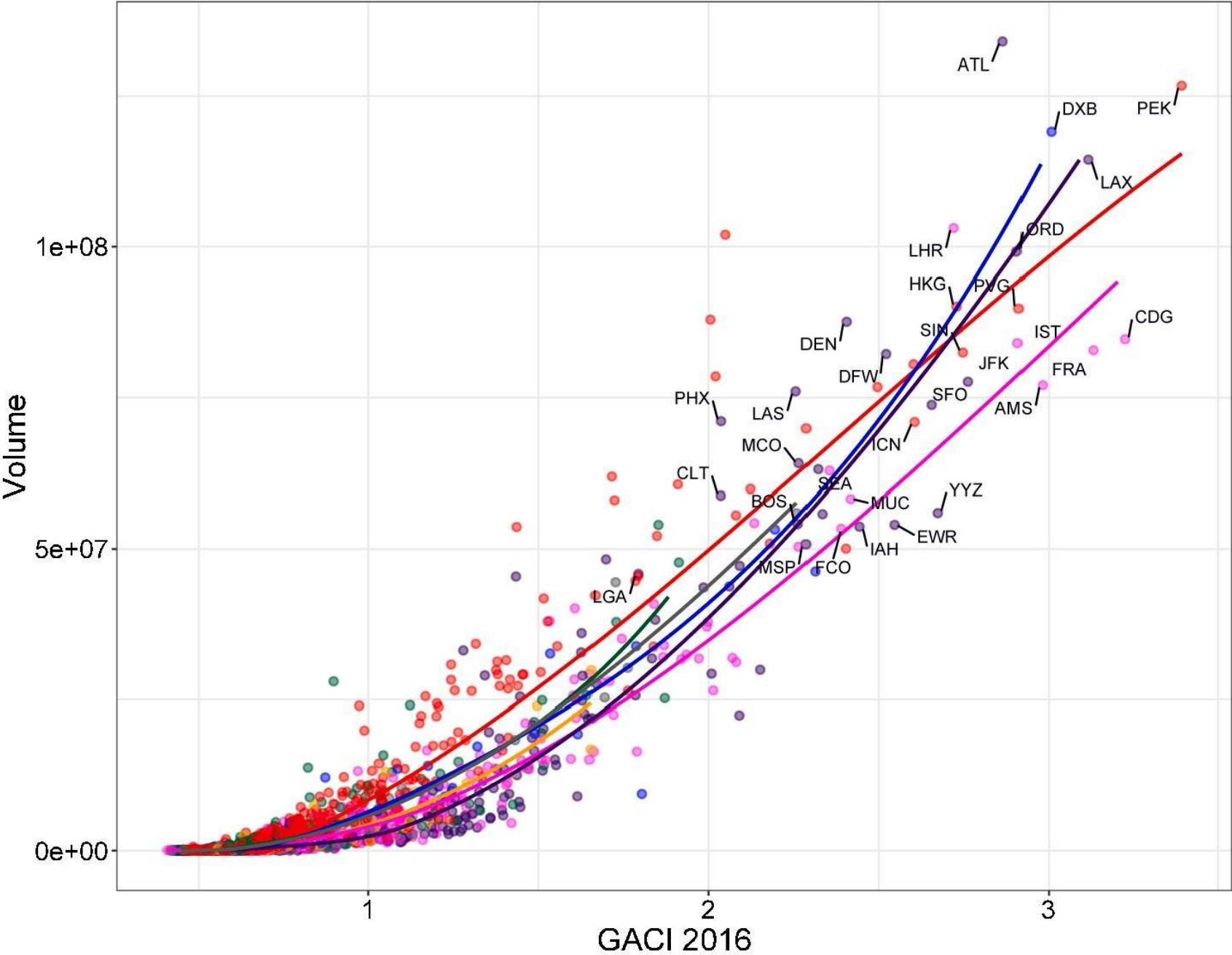
# Making GACI comparable over time

- A naive yearly PCA would re-scale every year, so scores could not be compared across years.
- **We standardise so rankings WITHIN a year and relative scores ACROSS years are both preserved.**
- Result: one index that tracks an airport or country up and down the global hierarchy from 1996 to 2023.

# Face validity: high GACI = the world's major hubs



*GACI tracks passenger volume; the top of the scale is ATL, PEK, DXB, LHR, HKG, SIN. Source: Cheung et al. (2020).*



## Step 3: from airports to countries, two ways

- **Hub quality = capacity-weighted MEAN (headline): average gateway quality, not inflated by having many tiny airports.**
- Total connectivity = SUM: overall network footprint, larger for big countries.
- We report both; hub quality is the headline because its first stage is stronger.

## Step 3: the aggregation formulas

### From airports to countries

*Two national summaries of the airport-level GACI*

**Hub quality (capacity-weighted mean) -- headline**

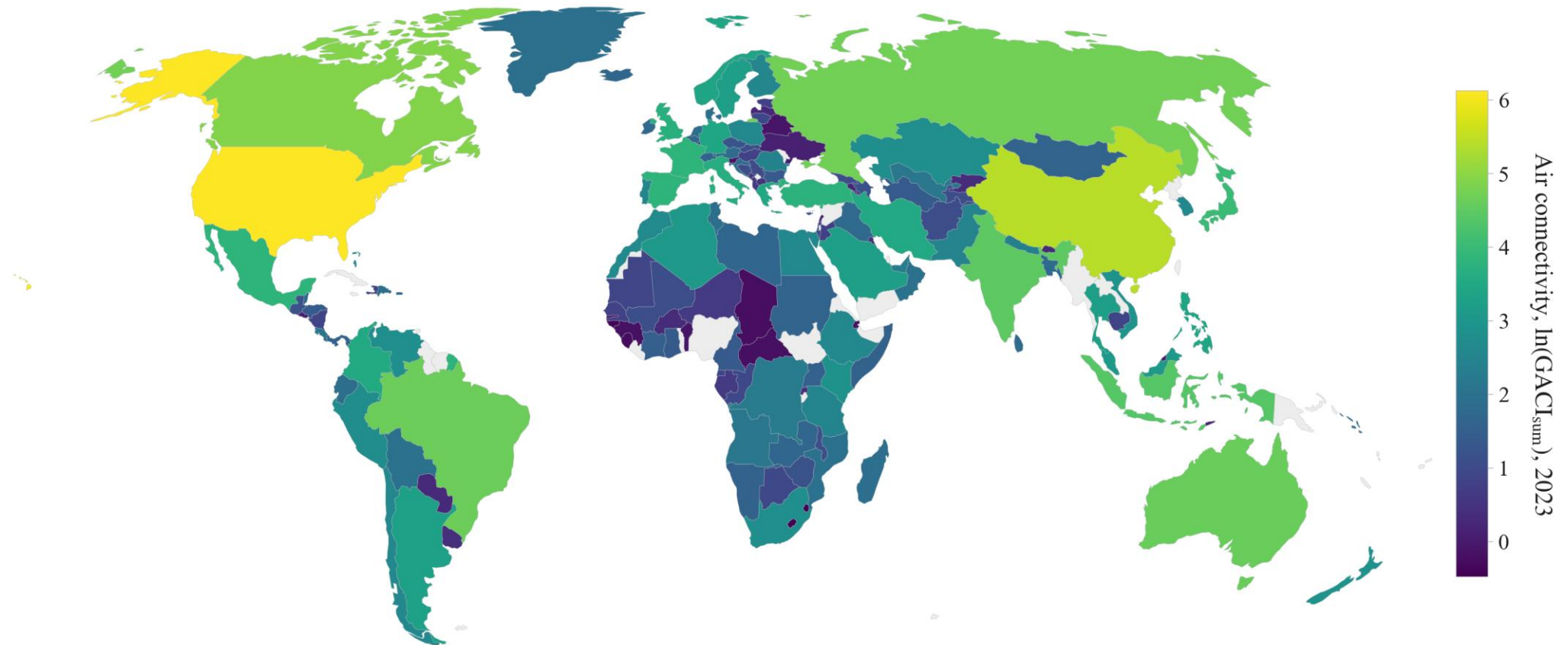
$$\ln \text{GACI}_c^{cwm} = \ln \left( \frac{\sum_{a \in c} \text{cap}_a \text{GACI}_a}{\sum_{a \in c} \text{cap}_a} \right)$$

**Total connectivity (country sum)**

$$\ln \text{GACI}_c^{sum} = \ln \left( \sum_{a \in c} \text{GACI}_a \right)$$

*Panel: 184 countries x 1996--2023 = 4{,}643 country-years*

# The built index, mapped: connectivity by country, 2023



Total connectivity (country sum), 2023. Yellow = highly connected. This is the explanatory variable for Part 03.



# 03

## Identification

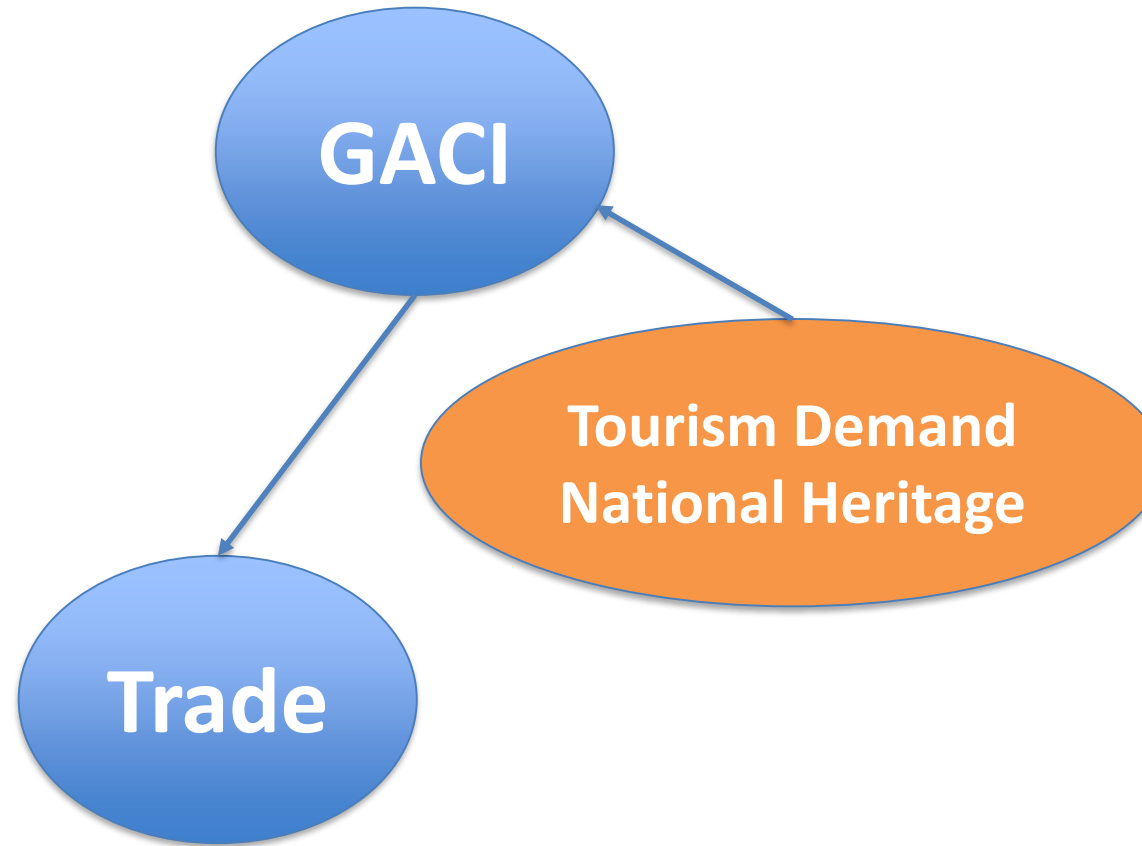
*Breaking reverse causality*

# The problem: connectivity is endogenous

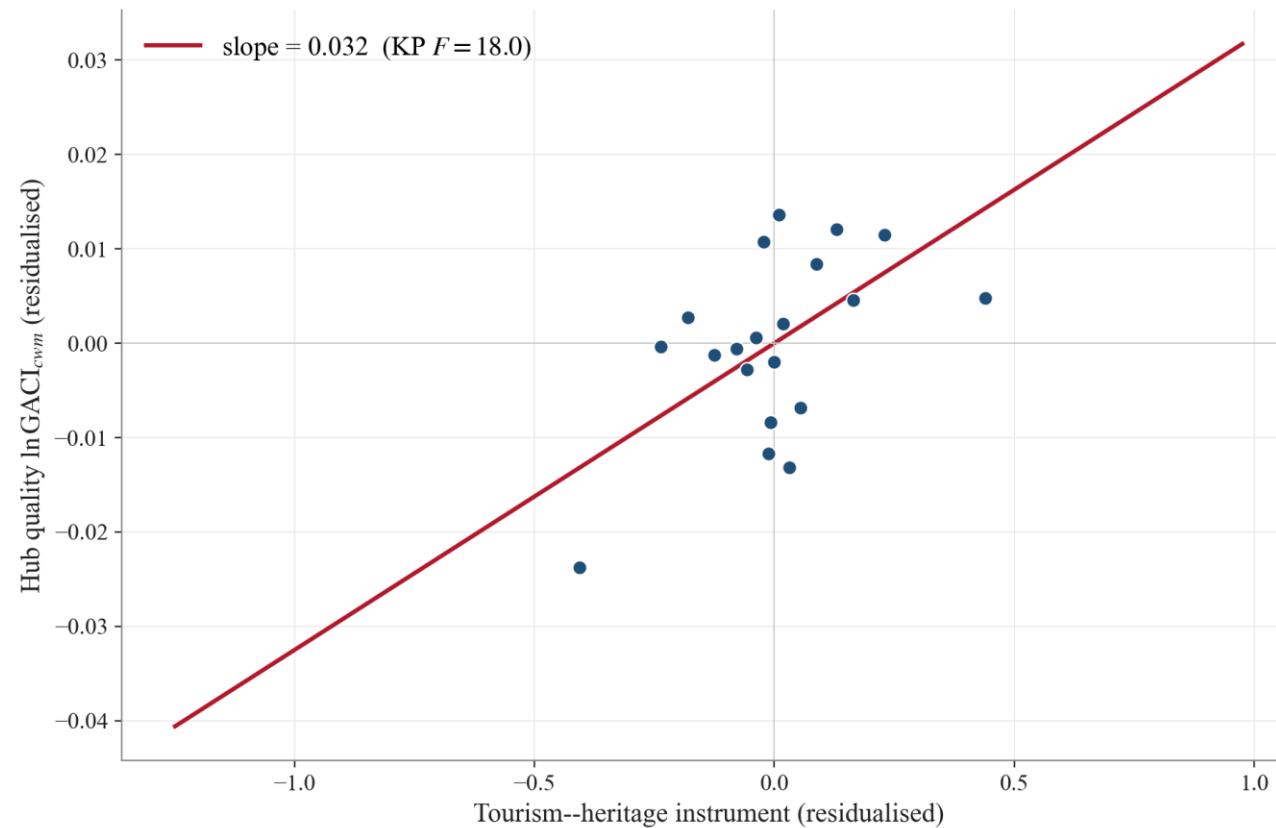
- Airlines add capacity WHERE TRADE IS ALREADY GROWING (reverse causality).
- Routes open lumpily: a coarse proxy for the connectivity that actually changes.
- OLS mixes the true effect with demand-driven entry and measurement error.

**We need variation in connectivity unrelated to goods-trade fundamentals.**

# A tourism–heritage shift-share instrument



# First stage: the instrument strongly predicts connectivity



*Binned partial-residual plot (country + year FE, population removed). Slope 0.032, Kleibergen–Paap F = 18.0.*

# 04

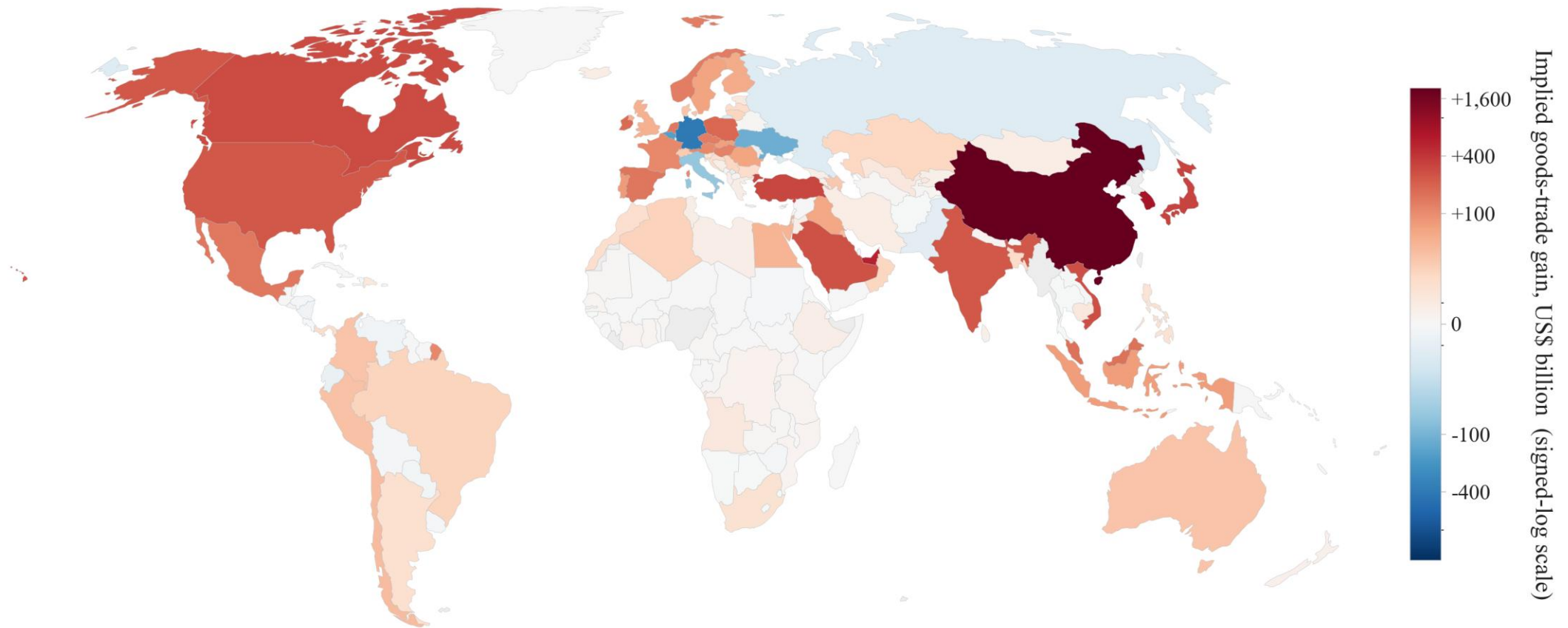
## Results

*Effect, who gains, and how much*

Table 2: Main results: OLS, first stage, and 2SLS, by connectivity measure (robust SE).

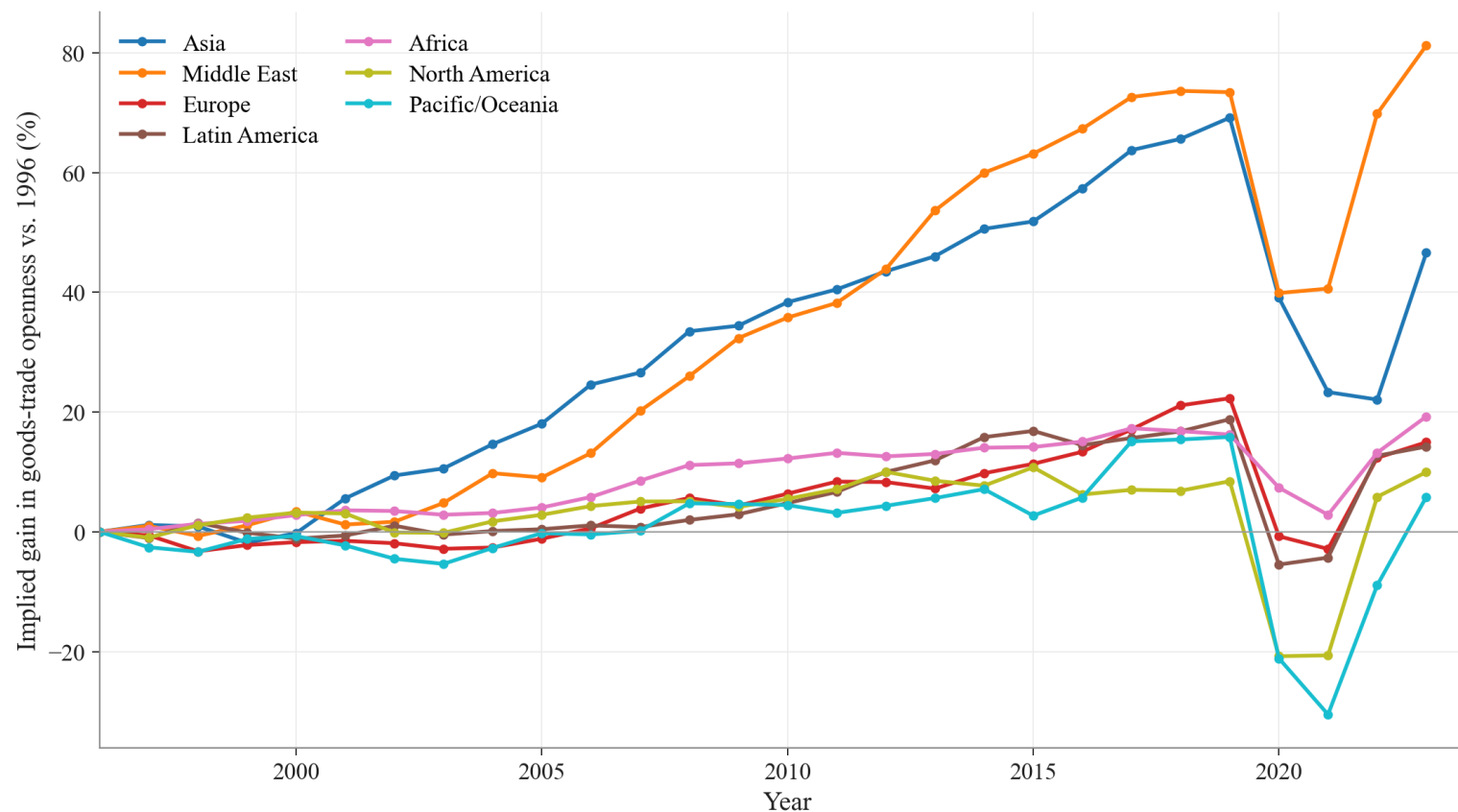
|                                                                                   | Trade openness<br>$\ln(g/Y)$    | Trade volume<br>$\ln g$ | GDP<br>$\ln Y$   |
|-----------------------------------------------------------------------------------|---------------------------------|-------------------------|------------------|
| <i>Panel A. Hub quality, <math>\ln \text{GACI}_{\text{cwm}}</math> (headline)</i> |                                 |                         |                  |
| First stage: instrument coef.                                                     | 0.032*** (0.007), KP $F = 18.0$ |                         |                  |
| OLS                                                                               | 0.157***                        | 1.144***                | 0.987***         |
| 2SLS                                                                              | 1.302**<br>(0.662)              | 2.303***<br>(0.776)     | 1.001<br>(0.680) |
| <i>Panel B. Total connectivity, <math>\ln \text{GACI}_{\text{sum}}</math></i>     |                                 |                         |                  |
| First stage: instrument coef.                                                     | 0.064*** (0.020), KP $F = 9.7$  |                         |                  |
| OLS                                                                               | -0.027                          | 0.237***                | 0.264***         |
| 2SLS                                                                              | 0.659*<br>(0.365)               | 1.166***<br>(0.432)     | 0.507<br>(0.341) |
| Country, Year FE                                                                  | Yes                             | Yes                     | Yes              |
| Observations                                                                      | 4,643                           | 4,643                   | 4,643            |

## Where the gains fall: ~\$7.8 trillion of 2023 trade



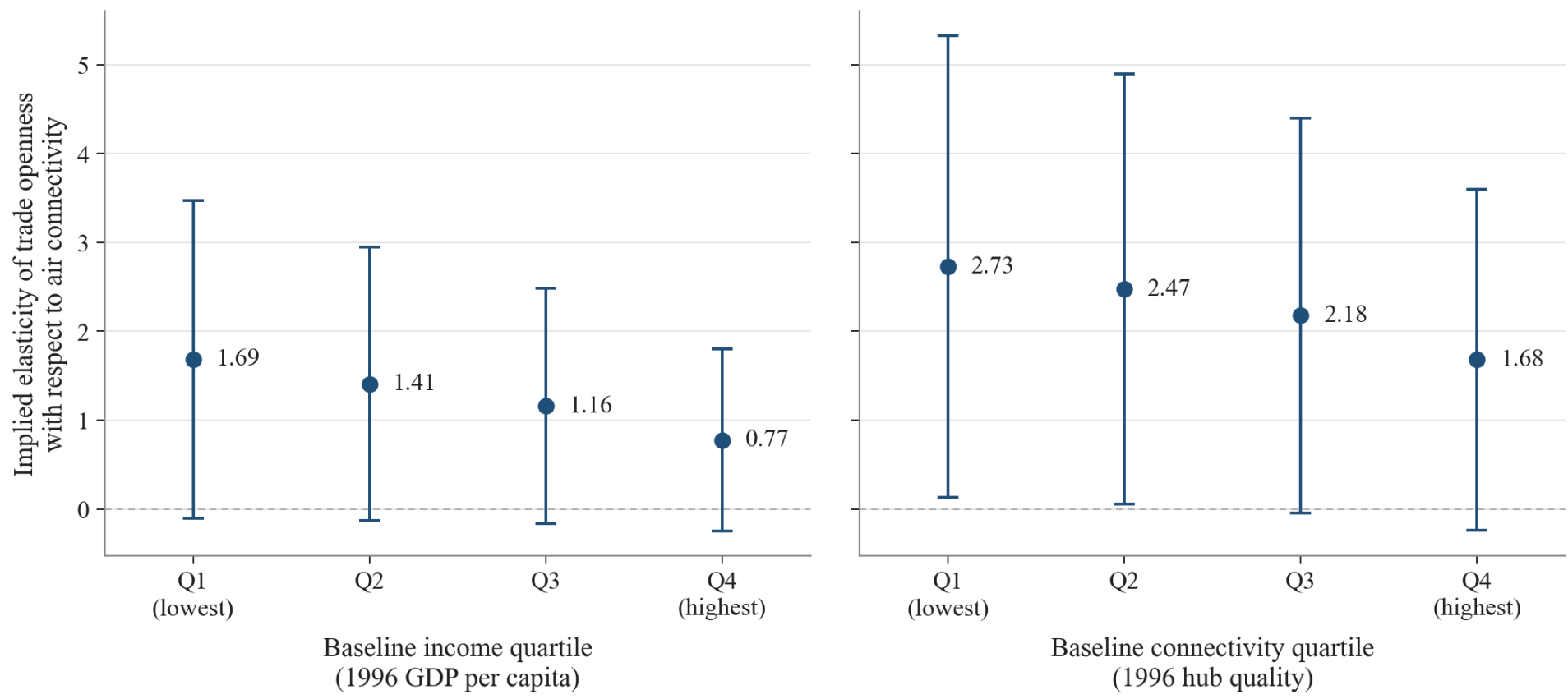
≈17% of world trade, 7.4% of world GDP. Concentrated in China, the Gulf, and East/Southeast Asia.

# When and where: Gulf and Asia pull away after 2005



*Implied gain in trade openness relative to 1996, by continent.*

# Who gains most: poorer, less-connected economies

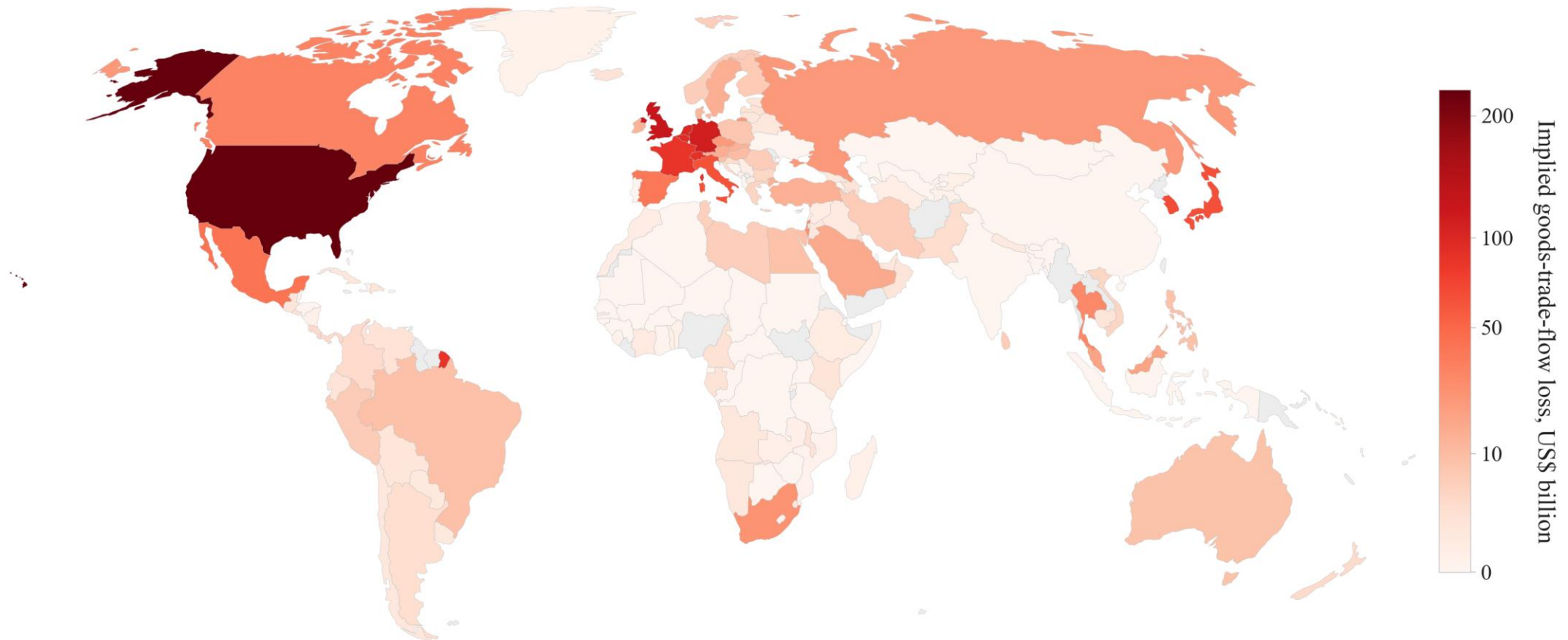


*Implied openness elasticity falls monotonically with baseline income and connectivity: aviation as a convergence lever.*

## Robustness: the design survives four checks

- Over-identification — heritage as two instruments: Hansen J not rejected ( $p = 0.52\text{--}0.84$ ); volume 2.18.
- Falsification — adding cultural heritage: Hansen J rejects ( $p = 0.004$ ).
- Independent instrument — Frankel–Romer/Feyrer geography: volume 2.26; combined, the two agree (J  $p = 0.95$ ).
- Timing — identification concentrated post-2010 (pre-2010 first stage  $F = 0.29$ ).

# Symmetry: the COVID-19 connectivity collapse



2019→2020: connectivity fell in 140 of 177 countries; implied trade disruption ~\$1.8 trillion, concentrated in major hubs.



# 05

## Conclusion

*What we learn, and what it implies*

## Summary and policy

- We build a 184-country, 1996–2023 panel of global aviation connectivity from the weighted world airport network (five centralities, PCA).
- Using a tourism–heritage IV, connectivity causally raises trade, mainly on the openness margin.
- Returns are largest for poorer, less-connected economies; large in aggregate (~\$7.8 trillion); symmetric in collapses (COVID-19).

**Airport/route investment and liberalisation: concrete trade levers, highest returns in peripheral economies.**

## Points to discuss

Why Trade?

How Trade? < Mechanism

# Thank you

*Global Air Connectivity and Trade Openness, 1996–2023*